



MME 321 Lab 4

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2/17/26

1.0 Introduction

The purpose of this lab was to model a coupled translational-rotational spring-mass-damper system. This allows the student to gain a broader understanding of how to model systems governed by translational and rotational subsystems as mathematical equations. During the course of this lab, the Simscape library within MATLAB's Simulink was used to compute and model the translational-rotational system in order to generate graphs of the system's dynamic inputs.

2.0 System Description

The coupled translational-rotational spring-mass-damper system is a physical system governed by the following parameters,

- $J = 1.0$
- $r = 0.4$
- $m = 0.5$
- $k = 1.0$
- $b = 0.25$
- $kr = 0.5$
- $br = 0.25$
- $F(t) = 1$

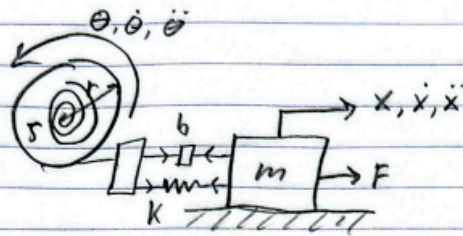
The system consists of a disk with moment of inertia J and radius r rotating connected to both a spring of force k with a constant of kr , and a damper of force b and constant of br . The other ends of the spring and damper are connected with a translational mass m that is being dragged away from the springs by a constant force $F(t)$. Due to the direction of motion of the mass m , the springs are in tension and the disk is being rotated counterclockwise.

3.0 Derivation of Equations of Motion

The derivation of the equations used to model the dynamic response of this mechanical system is shown in Fig.1. The system includes the variables: F_b , F_k , T_b , and T_k which are the damper force, spring force, rotational damper torque, and rotational spring torque respectively. One expression solves for x double dot and the other solves for θ double dot, which describes the translational and rotational motion of the system entirely.

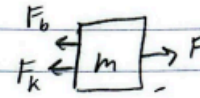
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Linear motion assume $x > r\theta$

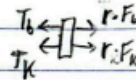
$$F_b = b(\dot{x} - r\dot{\theta}) \quad F_k = k(x - r\theta)$$



$$\sum F = m\ddot{x} \quad m\ddot{x} = F - F_b - F_k \Rightarrow \boxed{\ddot{x} = \frac{1}{m}(F - b(\dot{x} - r\dot{\theta}) - k(x - r\theta))} \quad \text{eq 1}$$

Rotational motion

$$T_b = b_r\dot{\theta} \quad T_k = k_r\theta$$



$$\sum T = J\ddot{\theta} \quad J\ddot{\theta} = r \cdot F_b + r \cdot F_k - T_b - T_k \Rightarrow \boxed{\ddot{\theta} = \frac{1}{J}(r(k(x - r\theta) + b(\dot{x} - r\dot{\theta}))) - b_r\dot{\theta} - k_r\theta} \quad \text{eq 2}$$

Fig.1

Furthermore the coupling relation used to convert rotational motion into translational motion can be described in the following equation:

$$x = r \cdot \theta$$

Where x , is a linear distance traveled, r is the radius of the disks, and θ is the rotational distance traveled by the disk.

During the previous lab, the exact same system was modeled using Simulink and MATLAB. The position, velocity, angular position, and force were then graphed and recorded in Fig.2. For the

purposes of this lab, these plots will be used as reference to the plots generated by Simscape in order to determine if Simscape has the same capabilities as MATLAB or Simulink.

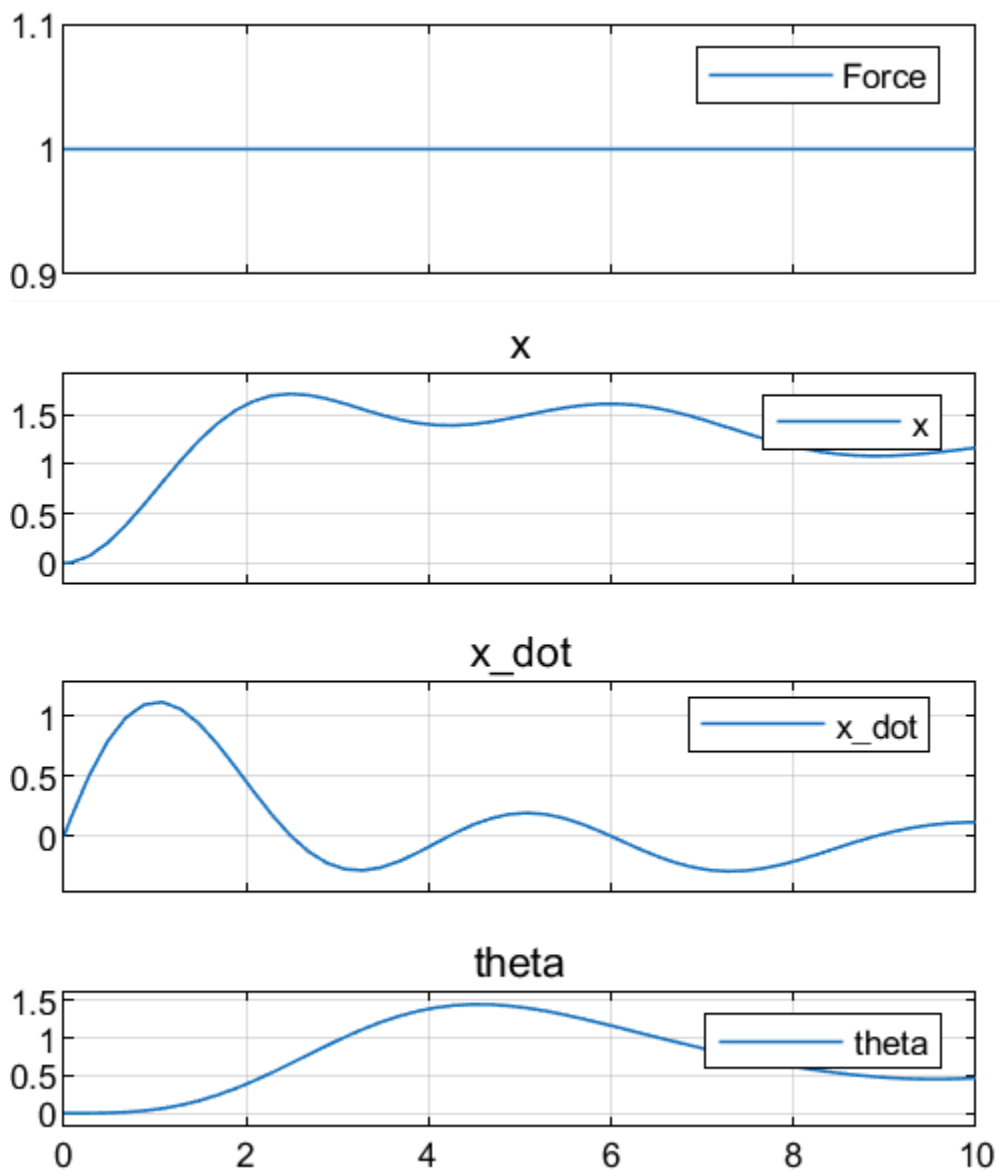


Fig.2

4.0 Simscape Modeling

Fig.3 Shows the entire system modeled using Simscape blocks within the Simulink package of MATLAB. The model includes named blocks that contain the parameter values defined by the specific system.

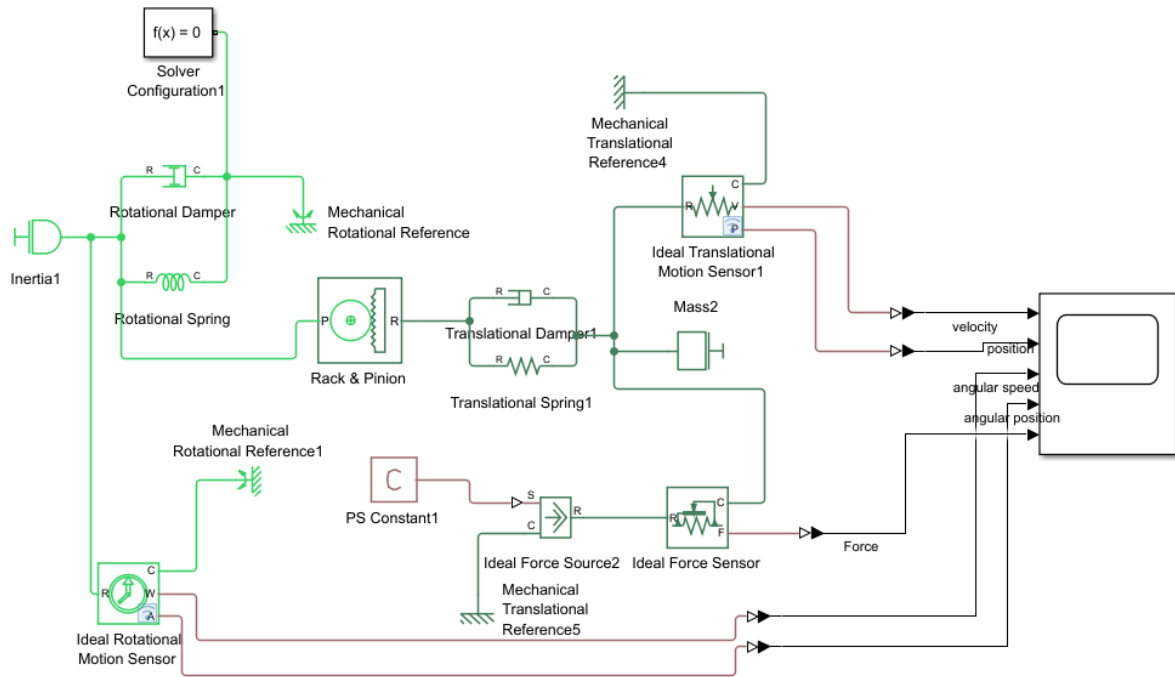


Fig.3

5.0 Simulation Results

Fig.4 shows dynamic outputs of the system generated by the Simscape model over a span of 10 seconds. This includes the overall translational position and velocity, rotational position and velocity, and constant applied force to the system.

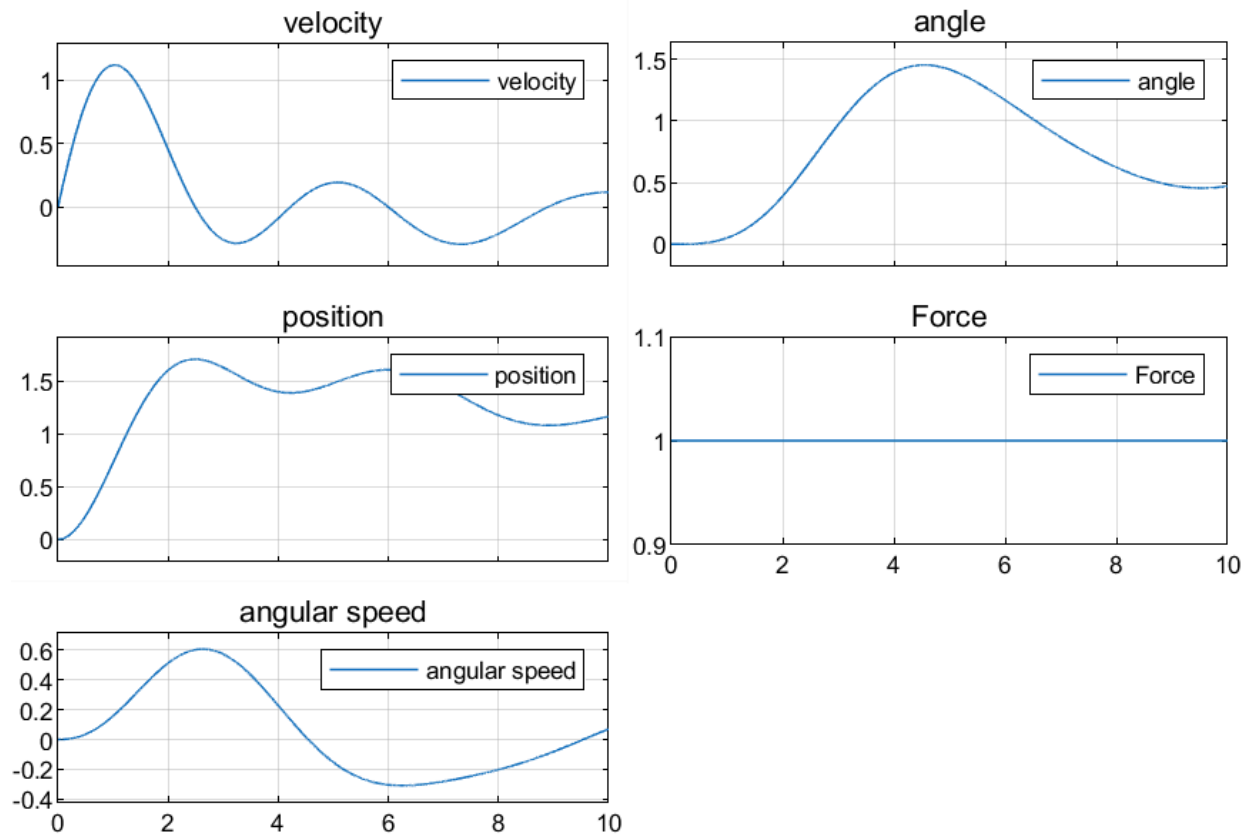


Fig.4

6.0 Conclusion

Overall, the plots generated by Simscape match the results of the previous lab where the same system was modeled in Simulink and MATLAB. The main difference between the Simscape and Simulink results is the aspect ratio of the graph windows, however the response is actually the exact same within the graphs for both methods. I found that personally Simscape takes a more nuanced understanding because you have to think about the physical system itself such as using a pinion and rack block to convert rotational movement to translational movement. That being said, with larger systems Simscape would be better for keeping track of subsystems and their outputs.